

SK Matrix-Model Contributions to the Squared Commutator

External assignment (by external-propagator appearance): $\phi_r^1(x_1) \phi_a^1(x_2) \phi_r^2(x_3) \phi_a^2(x_4)$, with $x_1 = (t, \mathbf{x})$, $x_2 = (0, \mathbf{0})$, $x_3 = (t, \mathbf{x})$, $x_4 = (0, \mathbf{0})$. Propagator notation: $G_{\ell_1 \ell_2}^{c_1 c_2}(z, y)$ (upper indices: time folds, lower indices: r/a labels). Each term carries explicit net matrix-size factor N_c^{p-4} (raw loop count p minus the overall $1/N^4$ normalization).

Topology 2

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.

Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.

Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\begin{aligned}
 \mathcal{T}_1 &= ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_1) G_{ra}^{11}(z_2, z_3) G_{rr}^{11}(x_1, z_1) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_2 &= ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_1) G_{ra}^{11}(z_2, z_3) G_{rr}^{11}(x_1, z_1) G_{ar}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_3 &= ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_3) G_{aa}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_4 &= 4ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_3) G_{ar}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_5 &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_3) G_{ra}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_6 &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_3) G_{ra}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{ar}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_7 &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_3) G_{ra}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_8 &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{aa}^{11}(z_2, z_3) G_{ra}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{ar}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_9 &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(x_1, z_1) G_{ra}^{11}(z_2, z_3) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{10} &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(x_1, z_1) G_{ra}^{11}(z_2, z_3) G_{ar}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{11} &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_3) G_{aa}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{12} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_3) G_{ar}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{13} &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{ra}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{aa}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{14} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{ra}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{ar}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{15} &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{ra}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{aa}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{16} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{ra}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{ar}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{17} &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_3) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(x_1, z_1) G_{ra}^{11}(z_2, z_1) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{18} &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_3) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(x_1, z_1) G_{ra}^{11}(z_2, z_1) G_{ar}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{19} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_3) G_{ar}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{20} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{aa}^{11}(z_2, z_3) G_{ar}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{ar}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{21} &= 4ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(z_2, z_3) G_{rr}^{11}(x_1, z_1) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{22} &= 4ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{ra}^{11}(z_2, z_3) G_{rr}^{11}(x_1, z_1) G_{ar}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{23} &= 4ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_3) G_{aa}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{24} &= 16ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_1) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_3) G_{ar}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{25} &= 2ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{ra}^{11}(x_1, z_1) G_{ra}^{11}(z_2, z_1) G_{aa}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{26} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{ra}^{11}(x_1, z_1) G_{ra}^{11}(z_2, z_1) G_{ar}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{27} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{aa}^{21}(x_4, z_3) G_{ra}^{21}(x_3, z_3) \\
 \mathcal{T}_{28} &= 32ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ar}^{11}(z_2, z_1) G_{ar}^{11}(z_2, z_3) G_{rr}^{11}(x_1, z_1) G_{rr}^{11}(z_2, z_1) G_{ar}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3) \\
 \mathcal{T}_{29} &= 8ig_2^3 N_c^{-4} G_{aa}^{11}(x_2, z_2) G_{aa}^{11}(z_1, z_3) G_{ra}^{11}(x_1, z_1) G_{ra}^{11}(z_2, z_1) G_{ra}^{11}(z_2, z_3) G_{rr}^{11}(z_2, z_1) G_{aa}^{21}(x_4, z_3) G_{rr}^{21}(x_3, z_3)
 \end{aligned}$$

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